

# Scopolamine (hyoscine): Drug information - UpToDate

## Scopolamine (hyoscine): Drug information - UpToDate

Scopolamine (hyoscine): Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Access UpToDate Lexidrug for additional drug information, tools, and databases.

Contributor Disclosures

For additional information see "Scopolamine (hyoscine): Patient drug information" and "Scopolamine (hyoscine): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Transdermal Scopolamine Safety Alert June 2025

The FDA has warned that Transderm Scop (scopolamine transdermal system) may cause hyperthermia, resulting in hospitalization or even death. Most cases have occurred in pediatric patients 17 years and younger and adults 60 years and older. The time from application of the patch to hyperthermia onset was typically less than 72 hours. The FDA is requiring updates to the prescribing information to include a warning and other information about this risk.

Further information may be found at <https://www.fda.gov/drugs/drug-safety-and-availability/fda-adds-warning-about-serious-risk-heat-related-complications-antinausea-patch-transderm-scop>.

Brand Names: US

- Transderm Scop;
- Transderm-Scop [DSC]

Brand Names: Canada

- ACCEL-Hyoscine;
- Buscopan

Pharmacologic Category

- Anticholinergic Agent

Dosing: Adult

### Dosage guidance:

**Dosing:** Each transdermal patch delivers ~1 mg scopolamine base over 3 days.

**Dosage form information:** In Canada, additional formulations include parenteral and oral scopolamine *butylbromide* (also known as hyoscine butylbromide) as well as parenteral scopolamine *hydrobromide* (also known as hyoscine hydrobromide). Scopolamine *butylbromide* and scopolamine *hydrobromide* formulations are not equivalent and are **not** interchangeable on a milligram-to-milligram basis.

**Clinical considerations:** Parenteral scopolamine *hydrobromide* is approved in Canada as an adjunct to anesthesia for sedation and amnesia; however, this use may no longer represent current clinical practice (Ref); refer to manufacturer's labeling for additional information. Scopolamine *butylbromide* salt is a quaternary derivative with fewer CNS effects compared with scopolamine *hydrobromide* and base (patch) formulations (Ref).

GI/GU spasm

#### **GI/GU spasm:**

**Note:** American College of Gastroenterology guidelines recommend against use of antispasmodics such as scopolamine for treatment of **global symptoms** of irritable bowel syndrome (IBS) (Ref).

*Scopolamine (hyoscine) butylbromide [Canadian product]:* **Oral:** 10 to 20 mg 3 to 5 times/day as needed; maximum: 60 mg/day (Ref). **Note:** When used for chronic symptom control, 10 mg 3 to 5 times/day as needed is recommended (Ref).

*Scopolamine (hyoscine) butylbromide [Canadian product]:* **IM/IV/SUBQ:** 10 to 20 mg once; if needed, may repeat at intervals of  $\geq 30$  minutes; maximum: 100 mg/day. **Note:** To avoid a potentially severe infusion reaction (ie, hypotension), a maximum rate of 20 mg/minute is preferred.

Malignant bowel obstruction, inoperable

#### **Malignant bowel obstruction, inoperable (off-label use):**

**Note:** Decreases gastric secretions, nausea, and vomiting; often given as part of a combination regimen (eg, with analgesic, antiemetic, or corticosteroid) (Ref).

*Scopolamine (hyoscine) butylbromide [Canadian product]:* **SUBQ :** Initial: 20 mg once followed by 60 mg over 24 hours as a continuous SUBQ infusion (Ref).

Motion sickness, prevention

#### **Motion sickness, prevention:**

*Scopolamine base:* **Transdermal:** Apply 1 patch (1 mg/3 days) behind ear 8 hours prior to required antiemetic effect for use up to 72 hours; if needed for >72 hours, remove old patch and place new one behind other ear (Ref). If symptoms are not adequately controlled with use of 1 patch, may consider using 2 patches (Ref).

Postoperative nausea and/or vomiting, prevention

#### **Postoperative nausea and/or vomiting, prevention:**

*Scopolamine base:* **Transdermal:** Apply 1 patch (1 mg/3 days) behind ear at least 1 to 2 hours prior to anesthesia or night before, and remove 24 hours after procedure (Ref).

Reduction of secretions at end of life

#### **Reduction of secretions at end of life (alternative agent) (off-label use):**

**Note:** Does not dry secretions already present and typically ineffective for lower respiratory tract secretions (Ref). Other agents are preferred over the transdermal patch and scopolamine hydrobromide due to CNS effects (eg, delirium, agitation) of scopolamine (Ref).

*Scopolamine base:* **Transdermal:** Apply 1 patch (1 mg/3 days) behind ear for up to 72 hours; if needed for >72 hours, remove old patch and place new one behind other ear (Ref).

*Scopolamine (hyoscine) butylbromide [Canadian product]:* Various regimens described; refer to local protocols. Example regimens include: **IV, SUBQ:** 20 mg every 4 to 6 hours as needed. Alternatively, administer 20 mg once followed by a continuous IV/SUBQ infusion of 20 to 60 mg over 24 hours (Ref). Maximum dose: 100 mg/day (Ref).

*Scopolamine (hyoscine) hydrobromide [Canadian product]:* **SUBQ:** 0.4 mg every 4 to 6 hours as needed **or** 0.4 mg followed by continuous infusion of 1.2 mg over 24 hours (Ref).

Sialorrhea

**Sialorrhea (alternative agent) (off-label use):**

**Note:** May be used to control excessive salivation associated with neuromuscular disorders (eg, cerebral palsy, amyotrophic lateral sclerosis [ALS]) (Ref).

*Scopolamine base*: **Transdermal:** Apply 1 patch (1 mg/3 days) behind ear; after 3 days, remove old patch and place new one behind other ear (Ref).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

**Altered kidney function:**

**Oral:** Scopolamine (hyoscine) *butylbromide* [Canadian product]: No dosage adjustment likely to be necessary due to limited systemic absorption (Ref).

**Parenteral:** Scopolamine (hyoscine) *butylbromide* [Canadian product]: There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied). Use with caution (~42% to 61% of a total dose is excreted in the urine) with frequent monitoring for adverse effects. If using for >48 hours in patients with eGFR <50 mL/minute/1.73 m<sup>2</sup>, consider reducing the dose or prolonging the dosing interval to prevent accumulation (Ref).

**Transdermal:** Scopolamine base: No dosage adjustment likely to be necessary (<10% of a total dose is excreted in the urine) (Ref). Use with caution and frequent monitoring in patients with severe kidney disease due to possible increased risk of CNS adverse effects (Ref).

**Hemodialysis, intermittent (thrice weekly):** Dialyzability unknown:

**Oral:** Scopolamine (hyoscine) *butylbromide* [Canadian product] : No dosage adjustment likely to be necessary due to limited systemic absorption (Ref).

**Parenteral:** Scopolamine (hyoscine) *butylbromide* [Canadian product] : There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution (~42% to 61% of a total dose is excreted in the urine) with frequent monitoring for adverse effects. If using for >48 hours, consider reducing the dose or prolonging the dosing interval to prevent accumulation (Ref).

**Transdermal:** Scopolamine base: No dosage adjustment likely to be necessary (<10% of a total dose is excreted in the urine) (Ref). Use with caution with frequent monitoring for adverse effects due to possible increased risk of CNS adverse effects (manufacturer's labeling); a case of ventricular tachycardia has also been reported in a patient with end-stage kidney disease (ESKD) and concomitant hepatic dysfunction (Ref).

**Peritoneal dialysis:** Dialyzability unknown:

**Oral:** Scopolamine (hyoscine) *butylbromide* [Canadian product] : No dosage adjustment likely to be necessary due to limited systemic absorption (Ref).

**Parenteral:** Scopolamine (hyoscine) *butylbromide* [Canadian product] : There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution (~42% to 61% of a total dose is excreted in the urine) with frequent monitoring for adverse effects. If using for >48 hours, consider reducing the dose or prolonging the dosing interval to prevent accumulation (Ref).

**Transdermal:** Scopolamine base: No dosage adjustment likely to be necessary (<10% of a total dose is excreted in the urine) (Ref). Use with caution with frequent monitoring for adverse effects due to possible increased risk of CNS adverse effects (Ref); a case of ventricular tachycardia has also been reported in a patient with ESKD and concomitant hepatic dysfunction (Ref).

**CRRT:**

**Oral:** Scopolamine (hyoscine) *butylbromide [Canadian product]* : No dosage adjustment likely to be necessary due to limited systemic absorption (Ref).

**Parenteral:** Scopolamine (hyoscine) *butylbromide [Canadian product]* : There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution with frequent monitoring for adverse effects. If using for >48 hours, consider reducing the dose or prolonging the dosing interval to prevent accumulation (Ref).

**Transdermal:** Scopolamine base: No dosage adjustment likely to be necessary (Ref).

**PIRRT (eg, sustained, low-efficiency diafiltration):**

**Oral:** Scopolamine (hyoscine) *butylbromide [Canadian product]* : No dosage adjustment likely to be necessary due to limited systemic absorption (Ref).

**Parenteral:** Scopolamine (hyoscine) *butylbromide [Canadian product]* : There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution with frequent monitoring for adverse effects. If using for >48 hours, consider reducing the dose or prolonging the dosing interval to prevent accumulation (Ref).

**Transdermal:** Scopolamine base: No dosage adjustment likely to be necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied). However, caution is recommended due to increased risks of adverse effects.

Dosing: Older Adult

Avoid use (Ref).

Dosing: Pediatric

(For additional information see "Scopolamine (hyoscine): Pediatric drug information")

**Dosage guidance:**

**Dosing:** Transdermal patch is designed to deliver 1 mg over 3 days. Do **not** cut patch to obtain smaller increments; remove only portions of the backing required for the dose (eg, if  $\frac{1}{2}$  patch is needed, remove only  $\frac{1}{2}$  of the backing prior to application), cover with occlusive dressing to keep in place (Ref).

Chronic drooling

**Chronic drooling:** Limited data available: Children  $\geq 3$  years and Adolescents: Transdermal: Initial: Apply  $\frac{1}{4}$  patch every 3 days for 1 week; may titrate to effect (control of secretions) by increasing by  $\frac{1}{4}$  patch every 7 days as tolerated. Maximum dose: 1 patch applied every 3 days (Ref).

Postoperative nausea and vomiting, prevention

**Postoperative nausea and vomiting, prevention:** Limited data available (Ref):

Children <2 years: Transdermal: Apply  $\frac{1}{4}$  patch several hours before surgery (as early as the night before surgery).

Children 2 to <6 years: Transdermal: Apply  $\frac{1}{2}$  several hours before surgery (as early as the night before surgery).

Children 6 to <12 years: Transdermal: Apply  $\frac{1}{2}$  to 1 patch several hours before surgery (as early as the night before surgery).

Children  $\geq 12$  years and Adolescents: Transdermal: Apply 1 patch several hours before surgery (as early as the night before surgery).

**Dosage adjustment for concomitant therapy:** Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

#### Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. However, caution is recommended due to increased risks of adverse effects.

#### Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. However, caution is recommended due to increased risks of adverse effects.

#### Adverse Reactions (Significant): Considerations

##### Anticholinergic effects

Scopolamine is associated with anticholinergic effects, including **urinary retention**, visual disturbances (eg, **blurred vision**, **mydriasis**), and **xerostomia**, in adult and pediatric patients that may contribute to discontinuation or nonadherence (Ref). **Hyperthermia**, including severe cases, has also been reported and may result in hospitalization or death; most cases occurred in patients  $\leq 17$  years of age and  $\geq 60$  years of age (Ref).

*Mechanism:* Dose-related; related to the pharmacologic action. Antagonism of muscarinic receptors (Ref).

*Onset:* Rapid; visual disturbances occurred within 1 to 2 days of patch application (Ref). Hyperthermia occurred within 3 days of patch application (Ref).

##### *Risk factors:*

- Concurrent medications that cause anticholinergic adverse reactions (eg, antihistamines, meclizine, tricyclic antidepressants, muscle relaxants)
- Older patients (Ref)
- Pediatric patients (Ref)

##### CNS effects

CNS effects, such as **drowsiness** and **fatigue**, may occur with scopolamine (Ref).

*Mechanism:* Related to the pharmacologic action. Directly related to the antagonism of muscarinic receptors (Ref).

##### *Risk factors:*

- Concurrent alcohol or medications that cause CNS effects (eg, sedatives, hypnotics, opiates, anxiolytics)
- Older patients (Ref)
- Pediatric patients (Ref)

##### Psychiatric effects

Scopolamine may cause psychiatric effects, including **acute psychosis** (ie, **delusion**, **hallucination**), **amnesia/memory impairment**, **agitation**, **confusion**, and **speech disturbance**, in adult and pediatric patients (Ref). Psychiatric effects are rare and reversible with discontinuation; may take 1 to 2 days to resolve following discontinuation (Ref).

*Mechanism:* Dose-related; related to the pharmacologic action. Directly related to the antagonism of muscarinic receptors (Ref).

*Onset:* Rapid; abnormal behaviors and psychosis may begin within 1 to 3 days of patch application (Ref).

##### *Risk factors:*

- Higher doses (Ref)
- Cognitive impairment or dementia (Ref)

- Older patients (Ref)
- Pediatric patients (Ref)

#### Withdrawal

Withdrawal symptoms following scopolamine discontinuation may include **diaphoresis, dizziness, drowsiness, fatigue, headache, hypotension, and nausea** in adult and pediatric patients (Ref). Resolution of symptoms occurs within several days to weeks (Ref).

*Mechanism:* Withdrawal; may be related to muscarinic receptor sensitization, leading to rebound overstimulation of the vestibular nuclei and reticular formation of the vomiting center with discontinuation (Ref).

*Onset:* Rapid; occurs within 18 to 72 hours after removal of the patch (Ref).

#### *Risk factors:*

- Prolonged use ( $\geq 3$  days) (Ref)

#### Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported with transdermal administration in adults.

>10%:

Gastrointestinal: Xerostomia (29% to 67%) (table 1)Xerostomia

Table 1: **Scopolamine: Adverse Reaction: Xerostomia**

| Drug (Scopolamine) | Placebo | Indication                                          | Number of Patients |
|--------------------|---------|-----------------------------------------------------|--------------------|
| 67%                | N/A     | Nausea and vomiting associated with motion sickness | N/A                |
| 29%                | 16%     | Post-operative nausea and vomiting                  | 461                |

Nervous system: Dizziness (12%) (table 2)Dizziness, drowsiness (8% to 17%) (table 3)Drowsiness

Table 2: **Scopolamine: Adverse Reaction: Dizziness**

| Drug (Scopolamine) | Placebo | Indication                         | Number of Patients (Scopolamine) |
|--------------------|---------|------------------------------------|----------------------------------|
| 12%                | 7%      | Post-operative nausea and vomiting | 461                              |

Table 3: **Scopolamine: Adverse Reaction: Drowsiness**

| Drug (Scopolamine) | Placebo | Indication                                          | Number of Patients |
|--------------------|---------|-----------------------------------------------------|--------------------|
| 17%                | N/A     | Nausea and vomiting associated with motion sickness | N/A                |
| 8%                 | 4%      | Post-operative nausea and vomiting                  | 461                |

1% to 10%:

Nervous system: Agitation (6%) (table 4)Agitation, confusion (4%) (table 5)Confusion

Table 4: **Scopolamine: Adverse Reaction: Agitation**

| <b>Drug (Scopolamine)</b> | <b>Placebo</b> | <b>Indication</b>                  | <b>Number of Patients (Scopolamine)</b> |
|---------------------------|----------------|------------------------------------|-----------------------------------------|
| 6%                        | 4%             | Post-operative nausea and vomiting | 461                                     |

Table 5: **Scopolamine: Adverse Reaction: Confusion**

| <b>Drug (Scopolamine)</b> | <b>Placebo</b> | <b>Indication</b>                  | <b>Number of Patients (Scopolamine)</b> |
|---------------------------|----------------|------------------------------------|-----------------------------------------|
| 4%                        | 3%             | Post-operative nausea and vomiting | 461                                     |

Ophthalmic: Mydriasis (4%) (table 6) Mydriasis, visual impairment (5%) (table 7) Visual Impairment

Table 6: **Scopolamine: Adverse Reaction: Mydriasis**

| <b>Drug (Scopolamine)</b> | <b>Placebo</b> | <b>Indication</b>                  | <b>Number of Patients (Scopolamine)</b> |
|---------------------------|----------------|------------------------------------|-----------------------------------------|
| 4%                        | 0%             | Post-operative nausea and vomiting | 461                                     |

Table 7: **Scopolamine: Adverse Reaction: Visual Impairment**

| <b>Drug (Scopolamine)</b> | <b>Placebo</b> | <b>Indication</b>                  | <b>Number of Patients (Scopolamine)</b> |
|---------------------------|----------------|------------------------------------|-----------------------------------------|
| 5%                        | 3%             | Post-operative nausea and vomiting | 461                                     |

Respiratory: Pharyngitis (3%)

Postmarketing:

Cardiovascular: Hypotension (associated with withdrawal) (Ref)

Dermatologic: Diaphoresis (associated with withdrawal) (Ref), erythema of skin, skin irritation, skin rash

Gastrointestinal: Nausea (associated with withdrawal) (Ref)

Genitourinary: Dysuria, urinary retention (Ref)

Local: Application-site reaction (including application-site blistering, application-site burning, application-site pruritus, application-site rash)

Nervous system: Acute psychosis (Ref), amnesia (Ref), ataxia (Ref), delirium (Ref), delusion (Ref), disorientation (Ref), disturbance in attention, fatigue (Ref), hallucination (Ref), headache (Ref), hyperthermia (Ref), insomnia (Ref), memory impairment (Ref), paranoid ideation, restlessness (Ref), seizure, speech disturbance, vertigo

Ophthalmic: Amblyopia, angle-closure glaucoma, blurred vision (Ref), dry eye syndrome, eye pruritus, eyelid pain (irritation), strabismus (Ref)

Contraindications

Injection:

Hyoscine butylbromide [Canadian product]: Hypersensitivity to hyoscine butylbromide, atropinics, or any component of the formulation; untreated narrow-angle glaucoma; megacolon, prostatic hypertrophy with urinary retention; stenotic lesions or mechanical stenosis of the GI tract; myasthenia

gravis; tachycardia, angina, heart failure; paralytic or obstructive ileus; IM administration in patients receiving anticoagulant therapy.

Scopolamine hydrobromide [Canadian product]: Hypersensitivity to scopolamine or any component of the formulation; glaucoma or predisposition to narrow-angle glaucoma; paralytic ileus; prostatic hypertrophy; pyloric obstruction; tachycardia secondary to cardiac insufficiency or thyrotoxicosis.

Oral [Canadian product]: Hypersensitivity to hyoscine butylbromide, atropinics, or any component of the formulation; glaucoma, megacolon, mechanical stenosis in the GI tract, myasthenia gravis, obstructive prostatic hypertrophy, paralytic or obstructive ileus.

Transdermal: Hypersensitivity to scopolamine, other belladonna alkaloids, or any component of the formulation; narrow-angle glaucoma

#### Warnings/Precautions

##### **Concerns related to adverse effects:**

- Anaphylaxis: Anaphylaxis including episodes of shock has been reported following parenteral administration; observe for signs/symptoms of hypersensitivity following parenteral administration. Patients with a history of allergies or asthma may be at increased risk of hypersensitivity reactions.
- Bradycardia (paradoxical): Lower doses (0.1mg) may have vagal mimetic effects (eg, increase vagal tone causing paradoxical bradycardia); these effects are likely mediated by blockade of muscarinic receptors at the level of the brain.

##### **Disease-related concerns:**

- Cardiovascular disease: Use with caution in patients with coronary artery disease, tachyarrhythmias, heart failure, or hypertension; evaluate tachycardia prior to administration.
- Gastrointestinal (GI) obstruction: Use with caution in patients with GI obstruction; when used for the treatment of smooth muscle spasm of the GI tract avoid continuous (daily) or prolonged use without evaluating source of abdominal pain. Patients should be instructed to report persistent or worsening abdominal pain with or without other symptoms (eg, nausea/vomiting, irregular bowel movements, bloody stool, hypotension).
- Genitourinary (GU) disease/obstruction: Use with caution in patients with GU obstruction, prostatic hyperplasia, or urinary retention; when used for the treatment of smooth muscle spasm of the GU tract, avoid continuous (daily) or prolonged use without evaluating source of the spasm.
- Glaucoma: Use transdermal product with caution in patients with open-angle glaucoma; may increase intraocular pressure; adjust glaucoma therapy as necessary.
- Hepatic impairment: Use with caution in patients with hepatic impairment.
- Hiatal hernia: Use with caution in patients with hiatal hernia with reflux esophagitis.
- Hyperthyroidism: Use caution in patients with hyperthyroidism; may have increased risk for arrhythmias.
- Parkinson disease: Adverse events (including dizziness, headache, nausea, vomiting) may occur following abrupt discontinuation in patients with Parkinson disease.
- Psychosis: Use with caution in patients with a history of psychosis; may exacerbate condition.
- Renal impairment: Use with caution in patients with renal impairment.
- Seizure disorders: Use with caution in patients with a history of seizure disorder; may exacerbate condition.
- Ulcerative colitis: Use with caution in patients with ulcerative colitis; may precipitate/aggravate toxic megacolon.

##### **Special populations:**



- Older patients: Use with caution in older patients; may be more susceptible to the anticholinergic, neurologic, and psychiatric adverse reactions of scopolamine. May be more susceptible of disruption in thermoregulation from the patch.
- Pediatric: Use with caution in pediatric patients; may be more susceptible to the anticholinergic, neurologic, and psychiatric adverse reactions of scopolamine.

***Dosage form specific issues:***

- Fructose: Tablets may contain sucrose; avoid use of tablets in patients who are fructose intolerant.
- Product interchangeability: Scopolamine (hyoscine) hydrobromide should not be interchanged with scopolamine butylbromide formulations; dosages are not equivalent.
- Transdermal patch: May contain conducting metal (eg, aluminum); remove patch prior to MRI.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Patch 72 Hour, Transdermal:

Transderm Scop: 1 MG/3DAYS (4 ea, 10 ea, 24 ea)

Transaerm-Scop: 1 MG/3DAYS (1 ea [DSC], 10 ea [DSC], 24 ea [DSC])

Generic: 1 MG/3DAYS (4 ea, 10 ea, 24 ea)

Generic Equivalent Available: US

Yes

Pricing: US

**Patch, 72-hour** (Scopolamine Transdermal)

1 mg/3days (per each): \$11.40 - \$24.28

**Patch, 72-hour** (Transderm Scop Transdermal)

1 mg/3days (per each): \$18.00

**Disclaimer:** A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as butylbromide:

Buscopan: 20 mg/mL (1 mL)

Generic: 20 mg/mL (1 mL)

Solution, Injection, as hydrobromide:

Generic: 0.4 mg/mL (1 mL); 0.6 mg/mL (1 mL)

Tablet, Oral, as butylbromide:

Buscopan: 10 mg [contains corn starch]

Generic: 10 mg

Administration: Adult

**Note:** Butylbromide [Canadian product] or hydrobromide [Canadian product] may be administered by IM, IV, or SubQ injection.

IM: **Butylbromide:** Intramuscular injections should be administered 10 to 15 minutes prior to radiological/diagnostic procedures.

IV:

**Butylbromide:** No dilution is necessary prior to injection; inject at a rate of 1 mL/minute

**Hydrobromide:** No dilution is necessary prior to injection.

Oral [Canadian product]: Tablet should be swallowed whole and taken with a full glass of water.

Transdermal: Apply to hairless area of skin behind the ear. Wear only one patch at a time; remove first patch and apply new patch behind the other ear. Wash hands before and after applying the patch to avoid drug contact with eyes. Do not cut the patch. Once patch has been affixed behind the ear, do not touch while being worn since pressure may cause scopolamine to release at the edge. Topical patch is programmed to deliver 1 mg over 3 days. For motion sickness prevention, apply 8 hours prior to required antiemetic effect; once applied, patch may remain applied for up to 72 hours. When used postoperatively for nausea/vomiting, the patch should be removed 24 hours after surgery. If patch becomes displaced, discard and apply a new patch. Once removed, fold the used patch in half with the sticky side together; dispose of used patches in the trash out of reach from children and pets.

Preparation for Administration: Adult

Injection:

IM: Butylbromide [Canadian product]: No dilution required.

IV:

Butylbromide [Canadian product]: No dilution is necessary prior to injection.

Hydrobromide: Dilute with an equal volume of sterile water.

Administration: Pediatric

Transdermal: Apply patch to hairless area behind 1 ear. Do not wear more than 1 patch at a time. Wash hands before and after application of disc to avoid drug contact with eyes. Transdermal patch is designed to deliver 1 mg over 3 days. Once patch has been affixed behind the ear, do not touch while being worn since pressure may cause scopolamine to release at the edge. Do not use any patch that has been damaged, cut, or manipulated in any way. If necessary to use smaller patch increments, do not cut patch; remove only portions of the backing required for the dose (eg, if  $\frac{1}{2}$  patch is needed, remove only  $\frac{1}{2}$  of the backing prior to application), cover with occlusive dressing to keep in place (Ref). If patch becomes displaced, discard and apply a new patch behind the other ear. Once removed, fold the used patch in half with the sticky side together; dispose of used patches in the trash out of reach from children and pets.

Storage/Stability

Injection:

Butylbromide [Canadian product]: Store at 15°C to 30°C (59°F to 86°F). Do not freeze. Protect from light and heat. Stable in D<sub>5</sub>W, D<sub>10</sub>W, NS, Ringer's solution, and LR for up to 8 hours.

Hydrobromide [Canadian product]: Store at 15°C to 30°C (59°F to 86°F). Protect from light. Do not freeze.

Tablet [Canadian product]: Store at 15°C to 30°C (59°F to 86°F). Protect from light and heat.

Transdermal system: Store at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

**Gastrointestinal/genitourinary spasm (scopolamine butylbromide [Canadian product]):** Treatment of abdominal spasms and pain and discomfort in the stomach, bowels, and biliary tract; prevention of spasm prior to radiological or diagnostic procedures. May also be used off label for management of pain associated with smooth muscle spasms in the urinary tract (Tytgat 2007).

**Motion sickness, prevention (scopolamine base):** Prevention of nausea and vomiting associated with motion sickness.

**Postoperative nausea and/or vomiting, prevention (scopolamine base):** Prevention of post-operative nausea and vomiting associated with recovery from anesthesia or opioid analgesia and surgery.

Use: Off-Label: Adult

Malignant bowel obstruction, inoperable; Reduction of secretions at end of life; Sialorrhea

Medication Safety Issues

Older Adult: High-Risk Medication:

Beers Criteria: Scopolamine is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to its highly anticholinergic properties and uncertain effectiveness as an antispasmodic (Beers Criteria [AGS 2023]).

Scopolamine is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) for lower urinary tract symptoms with benign prostatic hyperplasia (BPH) and high post-void residual volume, ie, >200 mL. Some disease states of concern include delirium, dementia, chronic cognitive impairment, narrow-angle glaucoma, overactive bladder, urge incontinence, and chronic constipation (O'Mahony 2023).

Other safety concerns:

Transdermal patch may contain conducting metal (eg, aluminum); remove patch prior to MRI.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

**Note:** Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acetylcholinesterase Inhibitors: May decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Aclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Agents with Clinically Relevant Anticholinergic Effects: May increase anticholinergic effects of Scopolamine. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amantadine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Benperidol. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benztropine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Benztropine. *Risk C: Monitor*

Biperiden: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Biperiden. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Bornaprine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bromperidol. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Cimetropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Cimetropium. *Risk X: Avoid*

CloZAPine: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of CloZAPine. Management: Consider alternatives to this combination whenever possible. If combined, monitor closely for signs and symptoms of gastrointestinal hypomotility and consider prophylactic laxative treatment. *Risk D: Consider Therapy Modification*

CNS Depressants: May increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Darifenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Darifenacin. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Dicyclomine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dicyclomine. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

DroNABinol: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of DroNABinol. *Risk X: Avoid*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Eluxadolone: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Eluxadolone. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fesoterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Fesoterodine. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Gastrointestinal Agents (Prokinetic): Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepotidacin: May decrease anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Glucagon: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Glycopyrrolate (Oral Inhalation): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Oral Inhalation). *Risk X: Avoid*

Glycopyrrolate (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Systemic). *Risk C: Monitor*

Glycopyrronium (Topical): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Hydroxyzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hyoscyamine: Anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects and Hyoscyamine may increase with coadministration. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Ipratropium (Nasal): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ipratropium (Oral Inhalation): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Itopride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Itopride. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levosulpiride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Levosulpiride. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Melperone: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methscopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methscopolamine. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirabegron: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Mirabegron. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nitroglycerin: Agents with Clinically Relevant Anticholinergic Effects may decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxatomide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

OxyBUTYnin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OxyBUTYnin. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Potassium Chloride: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Chloride. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium chloride. *Risk X: Avoid*

Potassium Citrate: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Citrate. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium citrate. *Risk X: Avoid*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pramlintide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. These effects are specific to the GI tract. *Risk X: Avoid*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propantheline: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Propantheline. *Risk C: Monitor*

Propiverine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

QuiNIDine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ramosetron: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Ramosetron. *Risk C: Monitor*

Revefenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Revefenacin. *Risk X: Avoid*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Rivastigmine: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Rivastigmine. Rivastigmine may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Management: Use of rivastigmine with an anticholinergic agent is not recommended unless clinically necessary. If the combination is necessary, monitor for reduced anticholinergic effects. *Risk D: Consider Therapy Modification*



Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Secretin: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Secretin. Management: Avoid concomitant use of anticholinergic agents and secretin. Discontinue anticholinergic agents at least 5 half-lives prior to administration of secretin. *Risk D: Consider Therapy Modification*

Sofpironium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Sofpironium. *Risk X: Avoid*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Agents with Clinically Relevant Anticholinergic Effects may increase serum concentrations of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Tiaprider: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Tiaprider. *Risk C: Monitor*

Tiotropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tiotropium. *Risk X: Avoid*

Tolterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tolterodine. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trospium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trospium. *Risk C: Monitor*

Umeclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

## Pregnancy Considerations

Scopolamine crosses the placenta.

When the transdermal patch is used to prevent nausea and vomiting associated with surgery, dose adjustments are required prior to cesarean delivery. Avoid use of transdermal patches in pregnant patients with severe preeclampsia; eclamptic seizures have been reported after IV and IM use.

## Breastfeeding Considerations

Scopolamine is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother. The Academy of Breastfeeding Medicine (ABM) states that scopolamine used as an antiemetic in the perioperative period is likely safe; however, because scopolamine may cause maternal sedation and adversely affect milk supply with repeated doses, other agents may be preferred (ABM [Reece-Stremtan 2017]).

## Monitoring Parameters

Body temperature, heart rate, urinary output, intraocular pressure, mental alertness.

## Mechanism of Action

Blocks the action of acetylcholine at parasympathetic sites in smooth muscle, secretory glands and the CNS; increases cardiac output, dries secretions, antagonizes histamine and serotonin; at usual recommended doses, causes blockade of muscarinic receptors at the cardiac SA-node and is parasympatholytic (ie, blocks vagal activity increasing heart rate)

## Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Butylbromide: Injection:  $\leq 15$  minutes.

Hydrobromide: Injection:  $\sim 15$  minutes (Renner 2005).

Scopolamine base: Transdermal: 6 to 8 hours.

Duration:

Hydrobromide: Injection:  $\sim 4$  hours.

Scopolamine base: Transdermal: 72 hours.

Absorption:

Butylbromide: Oral: Quaternary salts are poorly absorbed (local concentrations in the GI tract following oral dosing may be high).

Hydrobromide: IM, SubQ: Rapid.

Distribution:  $V_d$ : Butylbromide: 128 L.

Protein binding: Butylbromide:  $\sim 4\%$  (albumin).

Metabolism: Hepatic.

Bioavailability: Butylbromide: Oral: 8%.

Half-life elimination:

Butylbromide: Terminal: IV:  $\sim 5$  hours; Oral:  $\sim 6$  to 11 hours.

Hydrobromide: Injection:  $\sim 1$  to 3.5 hours (Renner 2005).

Scopolamine base: Transdermal: 9.5 hours.

Time to peak:

Butylbromide: Oral: ~2 hours.

Hydrobromide: IM: ~20 minutes; IV: ~5 minutes; SubQ: ~15 minutes (Renner 2005).

Scopolamine base: Transdermal: 24 hours.

Excretion:

Butylbromide: IV: Urine (42% to 61% [half as parent drug]), feces (28% to 37%).

Hydrobromide: Injection: Urine (variable; as parent drug and metabolites) (Renner 2005).

Scopolamine base: Transdermal: Urine (<10%, as parent drug and metabolites).

Patient Counseling Points

(For additional information see "Scopolamine (hyoscine): Patient drug information")

**What is this drug used for?**

- It is used to help motion sickness.
- It is used to treat GI (gastrointestinal) spasms.
- It is used to prevent upset stomach and throwing up from surgery.
- It is used during surgery.
- It may be given to you for other reasons. Talk with the doctor.

**All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:**

**All products:**

- Dry mouth
- Feeling dizzy or sleepy
- Diarrhea
- Upset stomach
- Sore throat
- Restlessness

**Skin patch:**

- Irritation where this drug was used

**WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:**

**All products:**

- Severe dizziness and passing out
- A fast heartbeat
- Trouble passing urine
- Change in eyesight, eye pain, or severe eye irritation
- Seeing halos or bright colors around lights
- Red eyes
- Larger pupils

- Seizures
- Trouble speaking
- Mental, mood, or behavior changes that are new or worse
- Feeling confused
- Hallucinations (seeing or hearing things that are not there)
- Memory problems or loss
- Shortness of breath
- Feeling hot
- Fever
- Not sweating during activities or in warm temperatures
- Severe stomach pain that lasts or gets worse with fever, upset stomach or throwing up, change in bowel movements, tender stomach, very bad dizziness or passing out, or blood in the stool
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

#### **Skin patch:**

- Dizziness, very upset stomach or throwing up, headache, problems with balance or walking, very bad dizziness or passing out, muscle weakness, or slow heartbeat after taking off the patch

**Note:** This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

**Consumer Information Use and Disclaimer:** This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (QA) Qatar: Antipan | Buscopan | Dividol | Hycom | Hyoban | Hyopan | Nospasm | Scopinal | Scopoderm | Spasmopan

Use of UpToDate is subject to the Terms of Use.

#### **REFERENCES**

1. 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc.* 2023;71(7):2052-2081. doi:10.1111/jgs.18372 [PubMed 37139824]

2. Al Jeraisy M, AlFuraih M, AlSaif R, AlKhalifah B, AlOtaibi H, Abolfotouh MA. Efficacy of scopolamine transdermal patch in children with sialorrhea in a pediatric tertiary care hospital. *BMC Pediatr.* 2020;20(1):437. [PubMed 32943036]
3. Apfel CC, Zhang K, George E, et al. Transdermal scopolamine for the prevention of postoperative nausea and vomiting: a systematic review and meta-analysis. *Clin Ther.* 2010;32(12):1987-2002. doi:10.1016/j.clinthera.2010.11.014 [PubMed 21118734]
4. Babin RW, Balkany TJ, Fee WE. Transdermal scopolamine in the treatment of acute vertigo. *Ann Otol Rhinol Laryngol.* 1984;93(1 Pt 1):25-7. doi:10.1177/000348948409300106 [PubMed 6703595]
5. Back IN, Jenkins K, Blower A, Beckhelling J. A study comparing hyoscine hydrobromide and glycopyrrolate in the treatment of death rattle. *Palliat Med.* 2001;15(4):329-336. doi:10.1191/026921601678320313 [PubMed 12054150]
6. Bar R, Gil A, Tal D. Safety of double-dose transdermal scopolamine. *Pharmacotherapy.* 2009;29(9):1082-1088. doi:10.1592/phco.29.9.1082 [PubMed 19698013]
7. Bennett M, Lucas V, Brennan M, Hughes A, O'Donnell V, Wee B; Association for Palliative Medicine's Science Committee. Using anti-muscarinic drugs in the management of death rattle: evidence-based guidelines for palliative care. *Palliat Med.* 2002;16(5):369-374. doi:10.1191/0269216302pm584oa [PubMed 12380654]
8. Brandt JC, Harman E, Winchester D. An unusual case of nonsustained ventricular tachycardia. *Ther Adv Cardiovasc Dis.* 2015;9(3):103-106. doi:10.1177/1753944715581137 [PubMed 25882751]
9. Brodtkorb E, Wyzocka-Bakowska MM, Lillevold PE, Sandvik L, Saunte C, Hestnes A. Transdermal scopolamine in drooling. *J Ment Defic Res.* 1988;32(pt 3):233-237. doi:10.1111/j.1365-2788.1988.tb01409.x [PubMed 3047394]
10. Buscopan (scopolamine) tablets and injection, solution [product monograph]. Laval, Quebec, Canada: Sanofi Consumer Health Inc; March 2021.
11. Cairncross JG. Scopolamine psychosis revisited. *Ann Neurol.* 1983;13(5):582. doi:10.1002/ana.410130524 [PubMed 6870212]
12. Chowdhury NA, Sewatsky ML, Kim H. Transdermal scopolamine withdrawal syndrome case report in the pediatric cerebral palsy population. *Am J Phys Med Rehabil.* 2017;96(8):e151-e154. doi:10.1097/PHM.0000000000000665 [PubMed 28081025]
13. Clissold SP, Heel RC. Transdermal hyoscine (Scopolamine). A preliminary review of its pharmacodynamic properties and therapeutic efficacy. *Drugs.* 1985;29(3):189-207. doi:10.2165/00003495-198529030-00001 [PubMed 3886352]
14. Curran HV, Schifano F, Lader M. Models of memory dysfunction? A comparison of the effects of scopolamine and lorazepam on memory, psychomotor performance and mood. *Psychopharmacology (Berl).* 1991;103(1):83-90. doi:10.1007/BF02244079 [PubMed 2006245]
15. Doyle E, Byers G, McNicol LR, Morton NS. Prevention of postoperative nausea and vomiting with transdermal hyoscine in children using patient-controlled analgesia. *Br J Anaesth.* 1994;72(1):72-76. [PubMed 8110556]
16. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
17. Frampton A, Spinks J. Hyperthermia associated with central anticholinergic syndrome caused by a transdermal hyoscine patch in a child with cerebral palsy. *Emerg Med J.* 2005;22(9):678-679. doi:10.1136/emj.2003.011254 [PubMed 16113205]
18. Gan TJ, Belani KG, Bergese S, et al. Fourth consensus guidelines for the management of postoperative nausea and vomiting. *Anesth Analg.* 2020;131(2):411-448. doi:10.1213/ANE.0000000000004833 [PubMed 32467512]

19. Good WV, Crain LS. Esotropia in a child treated with a scopolamine patch for drooling. *Pediatrics*. 1996;97(1):126-127. [PubMed 8545207]
20. Grant GJ, Reale S. Pharmacologic management of pain during labor and delivery. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed February 20, 2024.
21. Harman S, Walling AM. Palliative care: the last hours and days of life. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 5, 2025.
22. Horimoto Y, Tomie H, Hanzawa K, Nishida Y. Scopolamine patch reduces postoperative emesis in paediatric patients following strabismus surgery. *Can J Anaesth*. 1991;38(4, pt 1):441-444. [PubMed 2065410]
23. Hugel H, Ellershaw J, Gambles M. Respiratory tract secretions in the dying patient: a comparison between glycopyrronium and hyoscine hydrobromide. *J Palliat Med*. 2006;9(2):279-284. doi:10.1089/jpm.2006.9.279 [PubMed 16629557]
24. Hughes A, Wilcock A, Corcoran R, Lucas V, King A. Audit of three antimuscarinic drugs for managing retained secretions. *Palliat Med*. 2000;14(3):221-222. doi:10.1191/026921600670188257 [PubMed 10858832]
25. International Association for Hospice and Palliative Care (IAHPC). *The IAHPC Manual of Palliative Care*. 3rd ed. International Association for Hospice and Palliative Care; 2013. <https://hospicecare.com/what-we-do/publications/manual-of-palliative-care/>. Accessed August 5, 2021.
26. Kåss RM, Ellershaw J. Respiratory tract secretions in the dying patient: a retrospective study. *J Pain Symptom Manage*. 2003;26(4):897-902. doi:10.1016/s0885-3924(03)00292-6 [PubMed 14527758]
27. Khalif IL, Quigley EM, Makarchuk PA, Golovenko OV, Podmarenkova LF, Dzhanyayev YA. Interactions between symptoms and motor and visceral sensory responses of irritable bowel syndrome patients to spasmolytics (antispasmodics). *J Gastrointest Liver Dis*. 2009;18(1):17-22. [PubMed 19337628]
28. Knuf KM, Spaulding FM, Stevens GJ. Scopolamine toxicity in an elderly patient. *Mil Med*. 2019;184(11-12):937-938. doi:10.1093/milmed/usz086 [PubMed 31004425]
29. Lacy BE, Pimentel M, Brenner DM, et al. ACG clinical guideline: management of irritable bowel syndrome. *Am J Gastroenterol*. 2021;116(1):17-44. doi:10.14309/ajg.0000000000001036 [PubMed 33315591]
30. Lacy BE, Wang F, Bhowal S, Schaefer E; study group. On-demand hyoscine butylbromide for the treatment of self-reported functional cramping abdominal pain. *Scand J Gastroenterol*. 2013;48(8):926-935. doi:10.3109/00365521.2013.804117 [PubMed 23865591]
31. Lau SH, Vaneaton C. Scopolamine patch withdrawal syndrome. *Hosp Pharm*. 2014;49(3):218-220. doi:10.1310/hpj4903-218 [PubMed 24715737]
32. Lewis DW, Fontana C, Mehallick LK, Everett Y. Transdermal scopolamine for reduction of drooling in developmentally delayed children. *Dev Med Child Neurol*. 1994;36(6):484-486. [PubMed 7516297]
33. Lin CH, Lung HL, Li ST, Lin CY. Delirium after transdermal scopolamine patch in two children. *J Neuropsychiatry Clin Neurosci*. 2014;26(2):E01-E2. [PubMed 24763767]
34. MacEwan GW, Remick RA, Noone JA. Psychosis due to transdermally administered scopolamine. *CMAJ*. 1985;133(5):431-432. [PubMed 4027811]
35. Madariaga A, Lau J, Ghoshal A, et al. MASCC multidisciplinary evidence-based recommendations for the management of malignant bowel obstruction in advanced cancer. *Support Care Cancer*. 2022;30(6):4711-4728. doi:10.1007/s00520-022-06889-8 [PubMed 35274188]

36. Manno M, Di Renzo G, Bianco P, Sbordone C, De Matteis F. Unique scopolamine withdrawal syndrome after standard transdermal use. *Clin Neuropharmacol.* 2015;38(5):204-205. doi:10.1097/WNF.0000000000000099 [PubMed 26366965]
37. Mato A, Limeres J, Tomás I, et al. Management of drooling in disabled patients with scopolamine patches. *Br J Clin Pharmacol.* 2010;69(6):684-688. doi:10.1111/j.1365-2125.2010.03659.x [PubMed 20565460]
38. Mercadante S, Casuccio A, Mangione S. Medical treatment for inoperable malignant bowel obstruction: a qualitative systematic review. *J Pain Symptom Manage.* 2007;33(2):217-223. doi:10.1016/j.jpainsymman.2006.06.014 [PubMed 17280927]
39. Mercadante S, Ripamonti C, Casuccio A, Zecca E, Groff L. Comparison of octreotide and hyoscine butylbromide in controlling gastrointestinal symptoms due to malignant inoperable bowel obstruction. *Support Care Cancer.* 2000;8(3):188-191. doi:10.1007/s005200050283 [PubMed 10789958]
40. Minton JA, Tofade TS, Shah SA. Psychosis from anticholinergic medications administered at a smoking cessation clinic. *Journal of Pharmacy Practice.* 2009;22(5):489-493.
41. Mueller-Lissner S, Tytgat GN, Paulo LG, et al. Placebo- and paracetamol-controlled study on the efficacy and tolerability of hyoscine butylbromide in the treatment of patients with recurrent crampy abdominal pain. *Aliment Pharmacol Ther.* 2006;23(12):1741-1748. doi:10.1111/j.1365-2036.2006.02818.x [PubMed 16817918]
42. Nachum Z, Shupak A, Gordon CR. Transdermal scopolamine for prevention of motion sickness: clinical pharmacokinetics and therapeutic applications. *Clin Pharmacokinet.* 2006;45(6):543-566. doi:10.2165/00003088-200645060-00001 [PubMed 16719539]
43. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med.* 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]
44. Parr JR, Todhunter E, Pennington L, et al. Drooling reduction intervention randomised trial (DRI): comparing the efficacy and acceptability of hyoscine patches and glycopyrronium liquid on drooling in children with neurodisability. *Arch Dis Child.* 2018;103(4):371-376. [PubMed 29192000]
45. Patel PN, Ezzo DC. Withdrawal symptoms after discontinuation of transdermal scopolamine therapy: treatment with meclizine. *Am J Health Syst Pharm.* 2009;66(22):2024-2026. doi:10.2146/ajhp080569 [PubMed 19890085]
46. Peng X, Wang P, Li S, Zhang G, Hu S. Randomized clinical trial comparing octreotide and scopolamine butylbromide in symptom control of patients with inoperable bowel obstruction due to advanced ovarian cancer. *World J Surg Oncol.* 2015;13:50. doi:10.1186/s12957-015-0455-3 [PubMed 25889313]
47. Pergolizzi JV, Raffa R, Taylor R. Prophylaxis of postoperative nausea and vomiting in adolescent patients: a review with emphasis on combination of fixed-dose ondansetron and transdermal scopolamine. *J Drug Deliv.* 2011;2011:426813. [PubMed 21773046]
48. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine (ABM). ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med.* 2017;12(9):500-506.
49. Refer to manufacturer's labeling.
50. Reid SM, Westbury C, Guzys AT, Reddihough DS. Anticholinergic medications for reducing drooling in children with developmental disability. *Dev Med Child Neurol.* 2020;62(3):346-353. [PubMed 31495925]
51. Reinhart DJ, Klein KW, Schroff E. Transdermal scopolamine for the reduction of postoperative nausea in outpatient ear surgery: a double-blind, randomized study. *Anesth Analg.* 1994;79(2):281-284. doi:10.1213/00000539-199408000-00013 [PubMed 7639364]

52. Renner UD, Oertel R, Kirch W. Pharmacokinetics and pharmacodynamics in clinical use of scopolamine. *Ther Drug Monit.* 2005;27(5):655-665. doi:10.1097/01.ftd.0000168293.48226.57 [PubMed 16175141]
53. Rozzini R, Inzoli M, Trabucchi M. Delirium from transdermal scopolamine in an elderly woman. *JAMA.* 1988;260(4):478. [PubMed 3385907]
54. Safer DJ, Allen RP. The central effects of scopolamine in man. *Biol Psychiatry.* 1971;3(4):347-355. [PubMed 4950489]
55. Scopolamine hydrobromide injection [product monograph]. Montreal, Canada: Omega; May 2017.
56. Scopolamine transdermal system [prescribing information]. Pennington, NJ: Zydus Pharmaceuticals (USA) Inc; September 2024.
57. Sennhauser FH, Schwarz HP. Toxic psychosis from transdermal scopolamine in a child. *Lancet.* 1986;2(8514):1033. [PubMed 2877187]
58. Seo SW, Suh MK, Chin J, Na DL. Mental confusion associated with scopolamine patch in elderly with mild cognitive impairment (MCI). *Arch Gerontol Geriatr.* 2009;49(2):204-207. doi:10.1016/j.archger.2008.07.011 [PubMed 18834638]
59. Sunderland T, Tariot PN, Cohen RM, Weingartner H, Mueller EA 3rd, Murphy DL. Anticholinergic sensitivity in patients with dementia of the Alzheimer type and age-matched controls. A dose-response study. *Arch Gen Psychiatry.* 1987;44(5):418-426. doi:10.1001/archpsyc.1987.01800170032006 [PubMed 3579494]
60. Transderm Scop (scopolamine) [prescribing information]. Deerfield, IL: Baxter Healthcare; December 2023.
61. Transderm Scop (scopolamine) [prescribing information]. Deerfield, IL: Baxter Healthcare; June 2025.
62. Tytgat GN. Hyoscine butylbromide: a review of its use in the treatment of abdominal cramping and pain. *Drugs.* 2007;67(9):1343-1357. doi:10.2165/00003495-200767090-00007 [PubMed 17547475]
63. US Food and Drug Administration (FDA). FDA drug safety communication: FDA adds warning about serious risk of heat-related complications with antinausea patch Transderm Scop (scopolamine transdermal system). <https://www.fda.gov/drugs/drug-safety-and-availability/fda-adds-warning-about-serious-risk-heat-related-complications-antinausea-patch-transderm-scop>. Published June 18, 2025. Accessed June 27, 2025.
64. van Esch HJ, van Zuylen L, Geijteman ECT, et al. Effect of prophylactic subcutaneous scopolamine butylbromide on death rattle in patients at the end of life: The SILENCE randomized clinical trial. *JAMA.* 2021;326(13):1268-1276. doi:10.1001/jama.2021.14785 [PubMed 34609452]
65. Wilcock A, Howard P, Charlesworth S, eds. *Palliative Care Formulary.* 7th ed. Pharmaceutical Press; 2020. Accessed August 5, 2021.
66. Wildiers H, Dhaenekint C, Demeulenaere P, et al; Flemish Federation of Palliative Care. Atropine, hyoscine butylbromide, or scopolamine are equally effective for the treatment of death rattle in terminal care. *J Pain Symptom Manage.* 2009;38(1):124-133. doi:10.1016/j.jpainsymman.2008.07.007 [PubMed 19361952]
67. Wilkinson JA. Side effects of transdermal scopolamine. *J Emerg Med.* 1987;5(5):389-392. doi:10.1016/0736-4679(87)90144-2 [PubMed 3668203]

Topic 9915 Version 503.0

Source: UpToDate. Downloaded 2026-05-26.